Self-Supervised Learning with Recursive Quadrature Filters for Neural Time Series

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Abstract

We develop self-supervised methods for training Recursive Quadrature Filter (RQF) encoders on unlabeled time series. RQFs are biologically motivated, complex-valued temporal filters that form a special class of diagonal state-space models (SSMs). Each RQF acts as a band-pass filter whose selectivity is determined by two learnable parameters: its tuning frequency and bandwidth. We separately introduce a self-supervised learning objective based on prediction error; the encoder is trained so that its current state predicts its own future activity. In the *windowed* mode, the summary of the current window predicts the summary of a later window; in the *dense* mode, the state at the current timestep predicts future log-power. Each RQF encoder is pretrained on a dataset and then a readout (e.g., a task-specific classifier) is trained on the output of the RQF encoder, with the encoder either frozen or fully fine-tuned. For multichannel EEG and intracranial recordings, a shared encoder processes each channel separately, and a spatial readout combines the channel representations using electrode locations. Across four neural benchmarks using scalp or intracranial EEG, the pretrained RQF system performs close to the state of the art while using a small fraction of the parameters of the competing models. For example, a 110k-parameter RQF system achieves 0.9247 ± 0.0022 test AUROC on 100-s TUAB segments, above SAMBA (0.9054) at the same context with 9 times as many parameters, and 0.9040 ± 0.0019 on 10-s segments, comparable to EEG foundation models with 3.2M–369M parameters.

1 Introduction

Self-supervised pretraining has become the default route to general-purpose representations of physiological time series. The leading EEG encoders often follow the recipe established in language and vision: a learned tokenizer maps raw signal into patches or discrete codes, a Transformer is trained to predict masked patches or codes, and the encoder is scaled to millions or hundreds of millions of parameters [1–3]. The resulting systems perform well across benchmarks. However, because scale, learned input embeddings, and masked pretraining arrive together, it is difficult to tell how much of the result comes from capacity and how much from the recipe, and the encoder does not explicitly parameterize oscillatory dynamics. This leaves open whether a compact encoder whose dynamics match the statistical regularities of the input signals can achieve the same performance range without either scale or masking.

Here, we develop self-supervised learning with Recursive Quadrature Filters (RQFs), a class of biologically motivated complex-valued temporal filters [4, 5]. Each RQF learns a tuning frequency and a bandwidth and behaves as a band-pass filter. A stack of RQFs is therefore inherently suited

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to physiological signals such as EEG and intracranial recordings, whose information is carried by rhythms in distinct frequency bands. RQFs are *prospective* by construction; the recurrence evolves under a phase-rotation prior that extrapolates the oscillation it currently tracks (Section 2). In addition, the bottom-up input to each layer is itself prospective; the layer is driven by a first-order look-ahead of its input [4]. Both the phase-rotation prior and the prospective input enable the encoder to predict its immediate future, such that prediction error is a natural learning objective. Because RQFs are a special case of diagonal state-space models (SSMs), they can be trained with the same computationally-efficient parallel-scan algorithms as diagonal SSMs.

An RQF encoder learns by predicting its own future activity, i.e., based on statistical regularities in the input signals, with no task labels. The self-supervised objective is a loss on the encoder parameters that compares an earlier representation with one produced far later, at horizons of tens of milliseconds to seconds. A prediction head maps an earlier representation to the representation that the same encoder produces later. The *windowed* objective divides the state sequence into windows, summarizes each with parameter-free or learned pooling, and predicts the summary of a later window, whereas in the *dense* objective the state at each timestep predicts future log-power. The input-side prospective correction is a fixed first-order look-ahead of each layer's bottom-up drive that compensates for integration lag without adding learnable parameters [4]. The self-supervised objective trains the encoder, including frequencies and bandwidths of each RQF, together with the prediction head to predict future activity. Appendix C relates this objective to CPC, JEPA, and previous self-supervised state-space encoders.

To assess performance of RQFs with self-supervised prediction-error based learning, we designed experiments on four benchmarks that separate end-to-end system performance from representation quality. After pretraining, the encoder is reused for a supervised task either frozen or fully fine-tuned, under audited splits and validation-only model selection (Section 4). 1) TUAB tests whether the complete windowed system, pretrained only on the target corpus, performs within the foundation-model range, and a paired label-budget sweep measures when pretraining is most beneficial. 2) BrainTreebank compares the frozen and fully fine-tuned encoder. 3) ISRUC-SG1 tests the windowed system on five-class sleep staging [6]. 4) BCI-IV-2a tests cross-corpus transfer by using TUEG-pretrained weights for motor imagery, i.e., the pretraining and downstream corpora differ.

We find that small encoders perform surprisingly well. Across all four benchmarks the pretrained RQF system performs close to the state of the art with a small fraction of the parameters of the competing models (Figure 2). For example, with 110k parameters, windowed pretraining followed by full fine-tuning achieves 0.9247 ± 0.0022 test AUROC on 100-s TUAB segments, above SAMBA at the same context (0.9054, 1.0M parameters), and 0.9040 ± 0.0019 on 10-s segments, comparable to EEG foundation models with 3.2M–369M parameters. On intracranial BrainTreebank, we achieve 0.8822 ± 0.0011 AUROC with the pretrained RQF encoder frozen, 2.9 points above the best published model (BaRISTA) that has 3.4 times more parameters.

2 Background: Recursive Quadrature Filters

A *Recursive Quadrature Filter* (RQF) [4] is a scalar complex-valued temporal filter derived from an energy-based model of neural circuit function [7]. It receives an input drive $z(t) \in \mathbb{C}$, and its state $y(t) \in \mathbb{C}$ evolves as

$$\tau\, \dot{y} \;=\; -y + \gamma\, z + (1-\gamma)(1+\iota\omega\tau)\, y, \tag{1}$$

where $\tau > 0$ is a base time constant, fixed during pretraining (Appendix B); $\omega > 0$ is the learnable frequency parameter; and $\gamma \in (0,1)$ is the learnable bandwidth parameter. With $\alpha \triangleq \gamma/\tau$, $\omega_0 \triangleq (1-\gamma)\omega$, and $\lambda \triangleq -\alpha + \iota\omega_0$, Eq. (1) collapses to $\dot{y} = \lambda y + \alpha z$, a band-pass filter with unit peak gain at ω_0 and half-power bandwidth 2α that is stable for any $\gamma \in (0,1)$. The prior term $(1+\iota\omega\tau)\, y$ extrapolates the state by one time constant to first order under a phase-rotation model, making the recurrence prospective by construction. The complex state makes each scalar filter oscillatory. The state is complex-valued even when the input z is real, with its real and imaginary parts representing the output of a quadrature pair of causal, temporal bandpass filters (i.e., with the same frequency response amplitude but with a 90° phase shift at each frequency). Hence, the modulus of the output $|y|$ follows the amplitude envelope of the filter's band regardless of phase. In real coordinates, each RQF is a damped harmonic oscillator [4]. The phase-rotation prior gives the recurrence a model of how an oscillation continues, which a prediction objective can exploit.

An encoder stacks L layers of d such filters with learned parameters $\boldsymbol{\omega}^{(l)}, \boldsymbol{\gamma}^{(l)} \in \mathbb{R}^d$. The input to layer $l > 1$ is $\mathbf{z}_n^{(l)} = \mathbf{W}^{(l)} \rho\Big(\mathbf{y}_n^{(l-1)}\Big)$, where $\mathbf{W}^{(l)}$ is a complex inter-layer mixing matrix and ρ is an activation function (also called an output nonlinearity), e.g., split-ReLU applied to the real and imaginary components. The first layer receives the waveform at the dataset-specific sampling rate. In addition, we make the input to each layer prospective [4]: each layer is driven by $\mathbf{z}^{(l)} + \tau\, \dot{\mathbf{z}}^{(l)}$ in place of $\mathbf{z}^{(l)}$, a first-order look-ahead of its input. Under zero-order-hold discretization this becomes a two-tap update on the current and preceding input samples. These taps introduce no additional trainable parameters and leave the recurrent transition unchanged. Self-supervised pretraining optimizes the encoder, including its phase-prior frequencies and bandwidths, together with the prediction head to predict activity at longer horizons, from tens of milliseconds to seconds (Section 3).

The top-layer states $\mathbf{y}_n^{(L)} \in \mathbb{C}^d$ form the representation used during pretraining and by the downstream task readout. Because the RQF is a special case of a diagonal complex state-space model, the encoder trains with the same parallel-scan algorithms as modern SSMs. The encoders that we trained for the results reported below have 30k to 140k parameters (Appendix B).

3 Self-supervised learning with RQF encoders

The RQF encoder maps a raw sequence of T input samples to top-layer states $\mathbf{y}_1^{(L)}, \ldots, \mathbf{y}_T^{(L)}$ with $\mathbf{y}_n^{(L)} \in \mathbb{C}^d$. Self-supervised pretraining uses no task labels: the encoder is trained to predict its own future states from its current ones. We have tested two learning objectives. The *windowed* objective predicts between summaries of the state sequence, whereas in the *dense* objective the state at each timestep predicts future log-power. For both objectives, a source representation $\mathbf{s}_a$ is read from the states corresponding to an earlier time a; a learned *prediction head* f_ϕ maps it to an estimate $\hat{\mathbf{t}}_a = f_\phi(\mathbf{s}_a)$ of a target $\mathbf{t}_a^+$ corresponding to a later time; and the estimate is scored by cosine similarity, after ℓ_2 normalization, against the true target and against K negatives $\mathbf{t}_{a,k}^-$, e.g., corresponding to other times. With $\mathcal{A}$ the set of sources in a batch and β an inverse temperature, the loss is the InfoNCE objective [8]:

$$\mathcal{L}_{\mathrm{NCE}} \;=\; -\frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \log \frac{\exp(\beta\, \mathrm{sim}_a^+)}{\exp(\beta\, \mathrm{sim}_a^+) + \sum_{k=1}^{K} \exp(\beta\, \mathrm{sim}_{a,k}^-)}, \tag{2}$$

where sim_a^+ is the cosine similarity between $\hat{\mathbf{t}}_a$ and $\mathbf{t}_a^+$; $\mathrm{sim}_{a,k}^-$ is the cosine similarity between $\hat{\mathbf{t}}_a$ and the kth negative; and targets are detached. The prediction head is discarded after pretraining. The two objectives differ in the representations that are being used for prediction ($\mathbf{s}_a$ and $\mathbf{t}_a^+$), either temporal samples or abstracted summaries of RQF outputs, and in the temporal scale of the prediction horizon.

Windowed future prediction. The windowed objective predicts how the representation evolves at a coarser temporal scale. We divide the top-layer states into fixed, non-overlapping windows and summarize each one into a vector $\mathbf{t}_j$: the speech configuration uses a parameter-free mean, whereas the multichannel configurations use a learned pooler that combines the mean real and imaginary parts of RQF outputs in the window with its final state, followed by a linear projection and layer normalization. The source is the summary of window j and the target is the summary of window $j + h_w$, where the horizon h_w is an integer number of windows. Negatives are other window summaries in the batch, from the same input or from other inputs. A regression configuration instead minimizes the mean-squared error between the estimated and true future summaries and needs no negatives; the BrainTreebank diagnostic in Appendix A uses this regression loss.

Dense per-timestep prediction. In the dense objective, the state at the current timestep predicts future log-power. The source is the real and imaginary components of the state $\mathbf{y}_n^{(L)}$ at an anchor timestep n. The target is built from the states in a future interval of $N_w \geq 1$ samples that begins $\Delta \geq 1$ samples after the anchor, so Δ is the prediction horizon in samples. Specifically, the target is *future log-power*, capitalizing on the quadrature pair of filters to compute the demodulated mean power of the RQF outputs:

$$\mathbf{g}_n \;=\; \log\Big(\epsilon \,+\, \tfrac{1}{N_w} \textstyle\sum_{u=n+\Delta}^{n+\Delta+N_w-1} \big|\mathbf{y}_u^{(L)}\big|^2\Big), \tag{3}$$

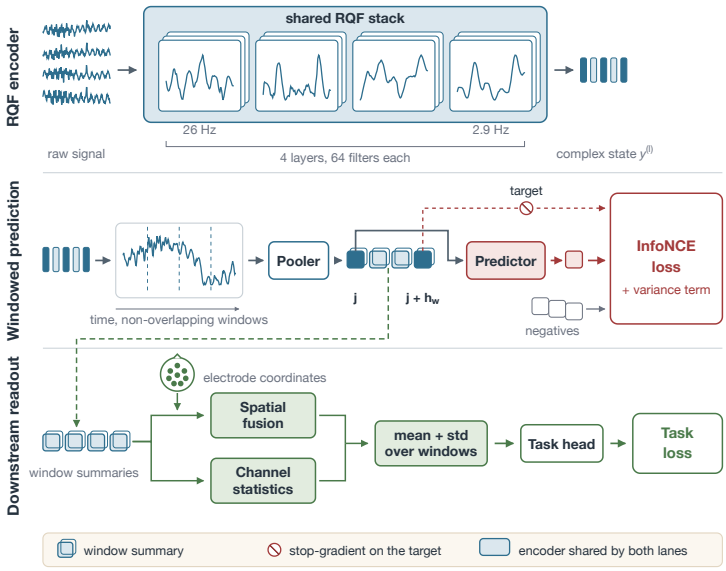


Figure 1: **Windowed self-supervised training and the multichannel task readout.** The three panels share one encoder. **Top:** A single RQF stack encodes every channel independently with four layers of 64 filters; the front plot of each layer shows the state magnitude of one filter, driven by a synthetic EEG-like probe input. The initialization ceiling of the frequency parameters halves from layer to layer, and the plotted exemplars run from 26 to 2.9 Hz. **Middle:** The state sequence is cut into non-overlapping windows, the learned pooler maps the mean and the real and imaginary parts of the final state of each window to a summary, and the predictor maps the summary of window j to the summary of window $j + h_w$. That later summary is the positive and enters with a stop-gradient, the negatives are other window summaries in the batch, and a variance term accompanies the InfoNCE objective. **Bottom:** The same summaries take two routes that are concatenated. One projects them and fuses them across channels (e.g., using the electrode coordinates), and the other takes their mean, standard deviation, and maximum across channels. The concatenation is pooled by its mean and standard deviation over windows before the task head (e.g., task-specific classifier). Dimensions, horizons, and optimization settings are in Table 11.

where u indexes temporal samples, with ϵ a small constant. Negatives are the log-power targets of other anchors.

Multichannel composition. Multichannel recordings add a spatial component without altering the temporal prediction task. During pretraining, the shared encoder processes each channel of each input as an independent single-channel sequence, and the pretext task receives neither electrode identity nor position. Spatial information enters only downstream, where a coordinate-aware readout combines the per-channel window summaries using the electrode coordinates. Figure 1 illustrates the windowed path and this readout. Appendix B gives the pooler and readout construction, anchor sampling, target calibration, the per-configuration negative exclusion policies, and the dataset-specific windows, horizons, and loss constants.

4 Evaluation design

The benchmarks address three questions: 1) whether the complete RQF system can match the performance range of much larger encoders; 2) when pretraining benefits a fixed downstream architecture; and 3) how temporal aggregation affects performance. TUAB tests performance with different amounts of labeled data, using patient-disjoint splits following BIOT [9]. Details of the test set are provided in Appendix A. BrainTreebank (BTB) tests whether the pretrained encoder supports sentence-onset detection [10]; we pretrain on BaRISTA's 17-session partition and evaluate using chronological splits within its seven held-out sessions [11]. ISRUC-SG1 tests the windowed learning

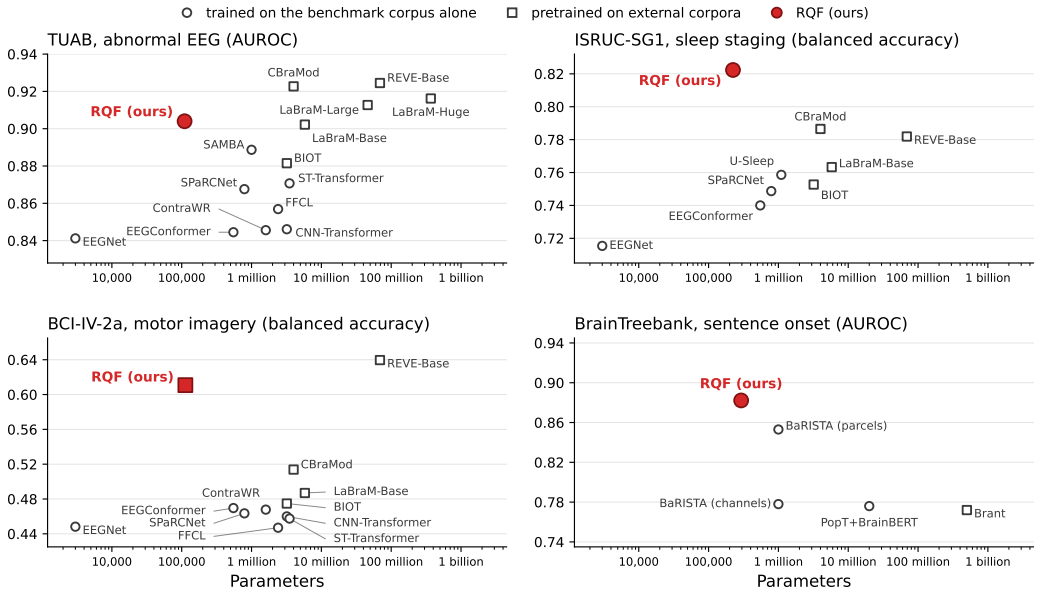


Figure 2: **Test performance versus parameter count on four benchmarks.** Each panel uses the metric named in its title. Red marks our strongest pretrained RQF configuration; circles denote training on the benchmark corpus and squares denote external-corpus pretraining. Published values are transcribed from their source papers, whose protocols differ from ours (Appendix B.2), so the panels give context, not controlled comparisons. All TUAB points, ours included, use 10-s segments; our 100-s result is compared with SAMBA at the same context in the text and Table 3. All BrainTreebank points use BaRISTA's chronological-split evaluation (its Appendix Table 16) [11]. Full values are in Appendix A.

objective on five-class sleep staging [6]. BCI-IV-2a tests cross-corpus transfer by pretraining the encoder on a subset of the TUH EEG corpus (TUEG) and testing with four-class motor imagery [12].

We compare each pretrained encoder with a control that starts from the same initial weights but skips pretraining (*paired init.*). During supervised training, the encoder is either kept fixed (*frozen*) or updated (*fully fine-tuned*, full-FT). For each benchmark, we use the test set only for final evaluation.

See Appendices A and B for full evaluation details including definitions of the performance metrics, and for a complete report of all of the results including supplementary results on a speech benchmark.

5 Results

Our results are of two kinds. Controlled comparisons fix the data, preprocessing, splits, and training settings and vary one factor (pretraining, freezing, objective, or label budget). Comparisons with published models use values from their source papers; these results give context but do not isolate a cause because the published protocols may differ from ours (Appendix B.2). When comparing model sizes, we count the real and imaginary parts of the complex-valued RQF parameters as 2 separate parameters.

A compact RQF compares favorably with large-scale models. On 100-s TUAB segments, windowed pretraining followed by full fine-tuning achieves 0.9247 ± 0.0022 test AUROC with 110,002 parameters, above SAMBA at the same context (0.9054, 9 times as many parameters). On the 10-s segments of the other published models, which Figure 2 plots, it achieves 0.9040 ± 0.0019, above LaBraM-Base (0.9022, 53 times as many). LaBraM-Large, LaBraM-Huge, CBraMod, and REVE-Base outperform it by 0.9, 1.2, 1.9, and 2.0 points with 418, 3,355, 36, and 627 times as many parameters (Figure 2; Tables 1 and 3).

Likewise, for BCI-IV-2a, full fine-tuning of TUEG-pretrained weights with Euclidean alignment achieves 0.6108 ± 0.0089 balanced accuracy, 9.7 points above CBraMod, which uses 35 times as

many parameters, and 2.9 points below REVE-Base, which uses 612 times as many (Figure 2 and Appendix Table 9). With this preprocessing, our RQF trained without SSL performs within 0.9 balanced-accuracy points of the versions pretrained on BCI-IV-2a or TUEG.

For ISRUC-SG1, the RQF system achieves 0.8223 ± 0.0028 balanced accuracy with 223,733 parameters, including the context module and classifier. It outperforms all of the comparison models, including REVE-Base, which uses 308 times as many parameters (Figure 2 and Appendix Table 5). Its paired initialization achieves 0.8252 ± 0.0016, so pretraining adds no measurable gain for this benchmark (95% interval over test subjects, -1.1 to 0.6 points). Both use context from 20 consecutive 30-second segments to predict the sleep stage of each.

Under full fine-tuning, pretraining helps most when labels are scarce. We use the TUAB benchmark to determine the benefit of pretraining with limited labeled data. We start two copies of the same encoder, both initialized with the same random weights, and we pretrain one of them with the windowed objective. We then add the readout and train both networks on the same labeled subset, allowing all weights (encoder + readout) to change. With only 1% of the training labels, the pretrained model achieves 0.7001 ± 0.0722 AUROC on 10-s segments, compared to 0.6509 ± 0.0230 without pretraining, a gain of 4.9 points (95% interval over test patients, 1.4 to 8.3), although one of three seeds is worse with pretraining. With all labels available, the two are level (0.9040 and 0.9046). Thus, self-supervised pretraining is most beneficial here when the supervised task provides little training data (Appendix Table 4).

The prediction objective shapes the learned representation. We use BrainTreebank to compare the two objectives. After pretraining, we keep the encoder frozen and train a task-specific readout. This tests whether the learned representation already contains information useful for the task. Windowed prediction (predicting window summaries) gives 0.8822 ± 0.0011 AUROC over five downstream seeds, 5.7 points above the same encoder frozen at paired initialization (0.8247 ± 0.0064; 95% interval over sessions, 2.8 to 9.4 points) and 1.5 points above dense prediction, which predicts future log-power from individual temporal samples (0.8670 ± 0.0055; 0.6 to 2.6 points; Appendix Table 7).

Updating the encoder when training the task-specific readout does not necessarily improve performance over keeping the encoder frozen. After windowed pretraining, the frozen encoder scores 1.2 points above full fine-tuning (0.8822 and 0.8706; 0.2 to 2.3 points), while paired initialization orders the other way (0.8247 frozen and 0.8459 fully fine-tuned; 0.4 to 4.1 points). These results suggest that on BrainTreebank, freezing the pretrained representation can be beneficial even when there is enough labeled data to update all weights.

6 Discussion, limitations, and conclusion

Three findings follow from the experiments. 1) A compact RQF encoder whose units learn a tuning frequency and a bandwidth is a viable alternative to Transformer-based models when the statistical regularities in the input signals are characterized by oscillations. On TUAB, for example, performance depends primarily on a few filters tuned for the alpha and beta frequency bands (Appendix Table 2). 2) The choice of source and target matters: predicting the summary of a later window from the summary of the current one yields stronger frozen representations than predicting future log-power from the state at the current timestep, although both lie well above the paired-initialization control. 3) Pretraining acts as a prior whose value depends on how the representation is used. The gain with pretraining is greatest when labels are scarce or the encoder is frozen. In our experiments, the gains are smaller under full fine-tuning, where the downstream loss is free to reshape the encoder.

The scope is limited in two ways. 1) Apart from the cross-corpus TUEG-to-BCI-IV-2a result, each corpus has its own pretrained weights, target geometry, and readout. 2) The comparisons with foundation models do not isolate the oscillatory bias from tokenization, masking, or scale.

In summary, future-state prediction is a practical way to pretrain RQF encoders on scalp EEG and intracranial recordings (also speech, see Appendix A). A small recurrent encoder learns competitive representations when its dynamics and its objective share a model of time, and RQFs provide that model for oscillatory signals.

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A Complete results by benchmark

This appendix reports the complete results behind the main-text comparisons. To measure the effect of SSL, we train a second model from the same starting weights without pretraining; tables label this control *paired init.* Published values use the evaluation procedures of their source papers. Relevant differences from our procedures are listed with each dataset in Appendix B.2.

Metrics. Binary tasks report AUROC, the probability that a positive example is scored higher than a negative one, and balanced accuracy (B-ACC), the mean recall across classes. ISRUC and BCI-IV-2a use B-ACC as their primary metric; Speech Commands (SC10) reports accuracy.

A.1 TUAB

TUAB contains clinical scalp EEG recordings from Temple University Hospital, labeled as normal or abnormal [13]. Our preparation contains 2,076 patients for training and validation and 253 for testing. We feed the EEG signals to the RQF encoder and compare the windowed and dense prediction objectives of Section 3, then train a readout to distinguish normal from abnormal recordings.

Full-data comparison. The windowed objective leads dense prediction more clearly in AUROC than in balanced accuracy. In a separate comparison on 100-s segments with both prediction horizons fixed at two seconds, the encoders frozen, and the default fine-tuning settings of Appendix B.4, the two objectives achieve 0.8978 ± 0.0089 and 0.8888 ± 0.0027 AUROC, respectively; their balanced accuracies differ by only 0.19 points, within the variation across seeds. The windowed results in Tables 1 and 3 use a one-second prediction horizon and the TUAB and ISRUC fine-tuning settings of Appendix B.4. The test set matches the official evaluation partition in size and class balance. The available metadata do not allow us to verify this correspondence patient by patient. Figure 3 shows how windowed pretraining retunes the filters of the TUAB encoder.

Table 1: TUAB per-segment test metrics on 10-s segments. Published values are from CBraMod, REVE, and SAMBA [2, 3, 14]; they use 200-Hz inputs except SAMBA, which uses 128 Hz as we do. Our results use three downstream seeds from a fixed pretraining checkpoint, and paired init. starts from the epoch-zero weights of that checkpoint without pretraining. Parameter counts follow the real-scalar convention of Table 10. Bold marks the configuration plotted in Figure 2; with full fine-tuning, paired init. is level with it.

Model	Params	Context	n	AUROC	B-ACC
EEGNet	3k	10 s/200 Hz	5	0.8412 ± .0031	0.7642 ± .0036
EEGConformer	550k	10 s/200 Hz	5	0.8445 ± .0038	0.7758 ± .0049
SPaRCNet	790k	10 s/200 Hz	5	0.8676 ± .0012	0.7896 ± .0018
ContraWR	1.6M	10 s/200 Hz	5	0.8456 ± .0074	0.7746 ± .0041
CNN-Transformer	3.2M	10 s/200 Hz	5	0.8461 ± .0013	0.7777 ± .0022
FFCL	2.4M	10 s/200 Hz	5	0.8569 ± .0051	0.7848 ± .0038
ST-Transformer	3.5M	10 s/200 Hz	5	0.8707 ± .0019	0.7966 ± .0023
BIOT	3.2M	10 s/200 Hz	5	0.8815 ± .0043	0.7959 ± .0057
LaBraM-Base	5.8M	10 s/200 Hz	5	0.9022 ± .0009	0.8140 ± .0019
LaBraM-Large	46M	10 s/200 Hz	5	0.9127 ± .0005	0.8226 ± .0015
LaBraM-Huge	369M	10 s/200 Hz	5	0.9162 ± .0016	0.8258 ± .0011
CBraMod	4.0M	10 s/200 Hz	5	0.9227 ± .0011	0.8289 ± .0022
REVE-Base	69M	10 s/200 Hz	n.r.	0.9245 ± .0013	0.8315 ± .0014
SAMBA-10s	1.0M	10 s/128 Hz	n.r.	0.8887 ± .0004	0.8150 ± .0037
Ours — paired init., frozen	110k	10 s/128 Hz	3	0.8643 ± .0009	0.7769 ± .0023
Ours — windowed SSL, frozen	110k	10 s/128 Hz	3	0.8939 ± .0010	0.8198 ± .0015
Ours — paired init., full-FT	110k	10 s/128 Hz	3	0.9046 ± .0017	0.8268 ± .0021
Ours — windowed SSL, full-FT	110k	10 s/128 Hz	3	**0.9040 ± .0019**	**0.8273 ± .0029**

Performance depends primarily on a few filters tuned for the alpha and beta frequency bands. On the TUAB validation set, we set to zero the states of all filters whose tuning frequency lies in one EEG band, in every layer of a trained model, and compare the AUROC drop with that of removing as

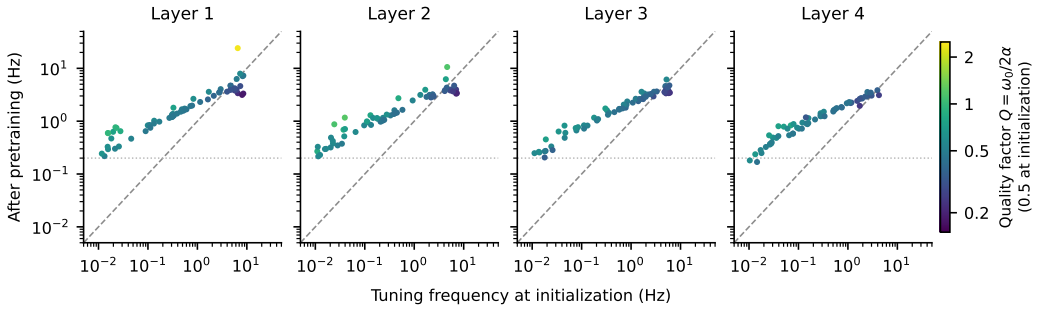


Figure 3: **Pretraining retunes the TUAB filters.** Tuning frequency $\omega_0/2\pi$ of each of the 64 filters per layer after windowed pretraining, against its value at initialization; the dashed line marks no change and the dotted line 0.2 Hz. Color gives the quality factor $Q = \omega_0/2\alpha$, which is 0.5 for every filter at initialization, the band-pass critical value [4]. Pretraining moves 246 of the 256 filters into 0.2 to 5 Hz; 98 start below 0.2 Hz and 2 remain there. In each layer, 31 to 50% of the filters end with Q above 0.5.

many randomly chosen filters from the same layers (Table 2). In the fully fine-tuned model, silencing its 5 to 7 alpha-tuned filters lowers AUROC by 13.4 points, against 0.5 for random filters, and its single beta-tuned filter costs 7.7 points. Filters tuned below 1 Hz matter less than random ones. Fine-tuning moves filters into the theta and alpha bands (28 to 33 and 5 to 7, against 16 and 1 in the frozen encoder). The models are not retrained after silencing, so the drops measure how much each model relies on these filters, not whether they are necessary.

Table 2: TUAB validation AUROC drop (points) when the states of all filters tuned to one band are set to zero in every layer, mean over three downstream seeds. Excess is the drop minus that of removing as many random filters from the same layers (20 draws); brackets are 95% intervals over patients (5,000 draws). No filter is tuned above 30 Hz.

	Windowed SSL, frozen			Windowed SSL, full-FT		
Band	Filters	Drop	Excess	Filters	Drop	Excess
Below 1 Hz	90	9.6	−7.7 [−9.2, −6.1]	87–89	4.2	−5.7 [−7.0, −4.5]
Delta (1–4 Hz)	148	32.2	9.6 [5.2, 14.1]	130–132	11.3	−5.7 [−7.2, −4.3]
Theta (4–8 Hz)	16	6.7	2.8 [1.5, 4.2]	28–33	7.1	2.2 [0.2, 4.2]
Alpha (8–13 Hz)	1	2.7	2.5 [1.2, 3.8]	5–7	13.4	12.9 [10.2, 15.8]
Beta (13–30 Hz)	1	3.1	3.0 [1.7, 4.4]	1	7.7	7.4 [5.7, 9.4]

100-s segments. With 100-s segments, the context of SAMBA-100s, windowed pretraining followed by full fine-tuning achieves 0.9247 ± 0.0022 test AUROC (Table 3). Its paired initialization achieves 0.9174 ± 0.0012, so pretraining adds 0.7 points (95% interval over test patients, 0.1 to 1.4). The same settings train on the whole segment without random cropping.

Table 3: TUAB per-segment test metrics on 100-s segments at 128 Hz, the configuration of SAMBA-100s [14]. Our results use three downstream seeds from the pretraining checkpoint of Table 1; both are fully fine-tuned, and paired init. starts from the same epoch-zero weights without pretraining. Bold marks our strongest configuration.

Model	Params	Context	n	AUROC	B-ACC
SAMBA-100s	1.0M	100 s/128 Hz	n.r.	$0.9054 \pm .0044$	$0.8264 \pm .0017$
Ours — paired init., full-FT	110k	100 s/128 Hz	3	$0.9174 \pm .0012$	$0.8357 \pm .0009$
Ours — windowed SSL, full-FT	110k	100 s/128 Hz	3	$\mathbf{0.9247 \pm .0022}$	$\mathbf{0.8426 \pm .0025}$

TUAB label efficiency. Pretraining helps most when only 1% of the training labels are available. With full fine-tuning, the benefit shrinks as labels increase and disappears with all labels. Frozen

pretrained encoders score 2.3 to 3.4 points above their paired-initialization controls at every label budget. Each comparison starts from the same weights and uses the same validation and test patients. Intervals over test patients exclude zero for full fine-tuning at 1% and for frozen encoders at 25% and 100%. At 1%, one of three seeds is worse with pretraining in both settings.

Table 4: TUAB label efficiency for windowed SSL and its paired initialization (test AUROC on 10-s segments; mean±std over three downstream seeds from a fixed checkpoint). All results use the 110k-parameter model of Table 10. Differences are computed from unrounded values. Bold marks the strongest configuration at each budget.

Configuration	1%	5%	10%	25%	100%
Full-FT, windowed SSL	**0.7001 ± .0722**	**0.8041 ± .0030**	**0.8395 ± .0054**	**0.8782 ± .0014**	0.9040 ± .0019
Full-FT, paired init.	0.6509 ± .0230	0.7822 ± .0035	0.8271 ± .0048	0.8736 ± .0020	**0.9046 ± .0017**
Full-FT difference (pp)	+4.91	+2.19	+1.24	+0.45	−0.06
Frozen, windowed SSL	0.6845 ± .0718	0.7906 ± .0069	0.8245 ± .0038	0.8603 ± .0013	0.8939 ± .0010
Frozen, paired init.	0.6501 ± .0060	0.7624 ± .0096	0.8014 ± .0017	0.8353 ± .0025	0.8643 ± .0009
Frozen difference (pp)	+3.44	+2.82	+2.32	+2.50	+2.96

A.2 ISRUC-SG1

ISRUC-Sleep contains overnight recordings collected at the Sleep Medicine Centre of Coimbra University Hospital; subgroup I has one recording from each of 100 adults undergoing sleep assessment [6]. We use six EEG channels from 99 participants, excluding one recording that lacks two required channels, with 79/10/10 participants for training, validation, and testing. We pretrain the RQF encoder with windowed future prediction (Section 3) and fine-tune it to assign one of five sleep stages to each 30-second epoch.

Epoch context. Adding neighboring epochs improves balanced accuracy from 0.7746 ± 0.0134 to 0.7990 ± 0.0087 when both models use the default fine-tuning settings of Appendix B.4. With the TUAB and ISRUC fine-tuning settings of the same appendix, the contextual model achieves 0.8223 ± 0.0028 (Table 5). Both RQF models first encode each 30-second epoch independently. The epoch-local model predicts from that representation alone; the contextual model passes the 20 representations through separate forward and reverse one-layer RQFs before predicting all 20 labels. The published baselines also use 20-epoch context, but CBraMod's evaluation combines the epoch representations with a shared one-layer Transformer [2]. Our contextual model uses RQFs for both epoch encoding and context, so both components differ from those baselines.

Table 5: ISRUC-SG1 five-class sleep-staging test balanced accuracy. Published baselines except REVE-Base come from CBraMod Table 7 and use its shared Transformer over 20 epochs [2]. Their parameter counts omit the Transformer, whereas ours include the complete downstream model. REVE reports 20-epoch sequence prediction without specifying its context module, so the table marks its context as reported [3]. Our split excludes subject 8, whose F3 and F4 channels are missing, so we train on 79 subjects instead of 80; validation and test subjects match CBraMod's. Our results are mean±std over three downstream seeds from a fixed pretraining checkpoint, and paired init. starts from the epoch-zero weights of that checkpoint without pretraining. Bold marks the configuration plotted in Figure 2, not the best value in the column.

Model	Params	Epoch context	B-ACC
EEGNet	3k	20×30 s	$0.7154 \pm .0121$
EEGConformer	550k	20×30 s	$0.7400 \pm .0133$
SPaRCNet	790k	20×30 s	$0.7487 \pm .0075$
ContraWR	1.6M	20×30 s	$0.7402 \pm .0126$
CNN-Transformer	3.2M	20×30 s	$0.7363 \pm .0087$
FFCL	2.4M	20×30 s	$0.7277 \pm .0182$
ST-Transformer	3.5M	20×30 s	$0.7381 \pm .0205$
DeepSleepNet	21M	20×30 s	$0.7419 \pm .0144$
U-Sleep	1.1M	20×30 s	$0.7586 \pm .0116$
BIOT	3.2M	20×30 s	$0.7527 \pm .0121$
LaBraM-Base	5.8M	20×30 s	$0.7633 \pm .0102$
CBraMod	4.0M	20×30 s	$0.7865 \pm .0110$
REVE-Base	69M	reported: 20×30 s	$0.7819 \pm .0078$
Ours — paired init., full-FT	224k	20×30 s	$0.8252 \pm .0016$
Ours — windowed SSL, full-FT	224k	20×30 s	$\mathbf{0.8223 \pm .0028}$

A.3 Speech Commands

Speech Commands is a collection of short spoken words recorded by volunteers, released by Google for keyword-recognition research [15]. We use the ten-command subset of 34,975 exactly one-second clips, with a class-stratified 70/15/15 training, validation, and test split. The RQF encoder receives the audio waveform and learns through the same windowed or dense prediction objective as in Section 3; the downstream task identifies the spoken command.

Dense and windowed prediction. On these non-neural signals, both objectives approach the strongest published compact models after full fine-tuning. With the encoder frozen, windowed prediction retains a 1.0-point advantage over dense prediction, while both remain far above their paired-initialization control (Table 6).

Table 6: SC10 ten-class test accuracy on the exact 34,975 one-second clips. Our models use raw waveforms and the same per-timestep readout. Our values are mean±std over three downstream seeds; bold marks our strongest configuration, not the best value in the column.

Model	Params	Input	n	Accuracy
Neural CDE [16]	89k	MFCC	5	$0.898 \pm .025$
CKConv [17]	100k	MFCC	n.r.	0.9527
FlexConv [18]	239k	MFCC	n.r.	0.9773
Ours — paired init., frozen	69.3k	raw	3	$0.6731 \pm .0043$
Ours — paired init., full-FT	69.3k	raw	3	$0.9666 \pm .0034$
Ours — dense SSL, frozen	69.3k	raw	3	$0.9408 \pm .0009$
Ours — windowed SSL, frozen	69.3k	raw	3	$0.9503 \pm .0012$
Ours — dense SSL, full-FT	69.3k	raw	3	$\mathbf{0.9723 \pm .0029}$
Ours — windowed SSL, full-FT	69.3k	raw	3	$0.9717 \pm .0010$

Phase-inclusive target diagnostic. In a separate SC10 diagnostic with an EMA target encoder, we compared power prediction alone with an additional phase-increment prediction head. Both frozen encoders were evaluated with the same diagnostic probe. The phase head achieved 0.885 validation

retrieval accuracy, but adding it reduced downstream validation macro recall from 0.8086 to 0.7045 with 5% of the labels and from 0.9040 to 0.8800 with all labels. This comparison covers one seed and does not test other ways of representing phase.

A.4 BrainTreebank

BrainTreebank records intracranial EEG from ten patients undergoing epilepsy monitoring at Boston Children's Hospital while watching movies [10]. We use 17 sessions for SSL and seven other sessions for downstream evaluation, with seven participants shared between these sets. We apply the windowed and dense objectives of Section 3 to the intracranial signals and train the downstream readout to detect sentence onsets. With the encoder frozen, windowed pretraining achieved 0.8822 ± 0.0011 AUROC, compared with 0.8247 ± 0.0064 for paired initialization (Table 7).

Evaluation protocol and published reference points. We use BaRISTA's chronological fold generator, which places validation blocks between training and test blocks, with five folds in each of its seven held-out sessions [11]. Published reference values come from BaRISTA's chronological evaluation (its Appendix Table 16), which uses the same sessions and folds. The evaluation sessions are held out from SSL, but each of the seven evaluation patients also contributes other sessions to pretraining.

Controlled comparisons. The controlled matrix separates the pretext objective from the way the encoder is used downstream. Table 7 uses the same architecture and seven sessions with five folds each while varying pretraining and whether the encoder is frozen. Table 8 tests the same transfer modes after regression pretraining; raw-voltage and STFT probes calibrate the task without a learned encoder. We retrain the controlled baselines on our folds.

Both non-neural baselines use L2-regularized logistic regression. The inputs are either standardized voltage samples or STFT magnitudes computed with a 512-sample Hann window, 384-sample overlap, and frequencies up to 150 Hz; regularization is selected on validation data.

Table 7: BrainTreebank sentence-onset test AUROC over seven held-out sessions with five chronological folds each. For each seed, we average the 35 session-by-fold results, then report mean±std across five downstream seeds. Published values are mean±s.e.m. from BaRISTA's chronological evaluation. Bold marks our strongest configuration, not the best value in the column.

Model	Params	Seeds	AUROC
Brant	500M	5	0.772 ± .009
PopT+BrainBERT	20M	5	0.776 ± .009
BaRISTA (channels)	1.0M	5	0.778 ± .009
BaRISTA (parcels)	1.0M	5	0.853 ± .007
Ours — paired init., frozen	293k	5	0.8247 ± .0064
Ours — paired init., full-FT	293k	5	0.8459 ± .0077
Ours — dense InfoNCE SSL, frozen	293k	5	0.8670 ± .0055
Ours — dense InfoNCE SSL, full-FT	293k	5	0.8529 ± .0042
Ours — windowed InfoNCE SSL, frozen	293k	5	**0.8822 ± .0011**
Ours — windowed InfoNCE SSL, full-FT	293k	5	0.8706 ± .0022

Table 8: Controlled BrainTreebank baselines and windowed-MSE diagnostic across the same seven sessions with five folds each (test AUROC). Logistic results are deterministic fits, mean±std across these 35 results; MSE results are mean±std over five downstream seeds.

Configuration	Test AUROC
Logistic regression, raw voltage	0.7633 ± .1091
Logistic regression, STFT magnitude	0.8621 ± .0807
Windowed MSE SSL, frozen	0.8728 ± .0043
Windowed MSE SSL, full-FT	0.8727 ± .0030

Six contrasts were fixed before the test set was evaluated, each paired over the 35 session-by-fold results after averaging the five downstream seeds. Their 95% intervals resample the seven sessions

with replacement, keeping each session's folds together (5,000 draws). Frozen windowed SSL exceeds its fully fine-tuned version by 1.2 points (interval 0.2 to 2.3), frozen dense SSL by 1.5 points (0.6 to 2.6), the STFT probe by 2.0 points (0.04 to 4.2), and frozen paired initialization by 5.7 points (2.8 to 9.4). Paired initialization orders the other way, with full fine-tuning 2.1 points above its frozen version (0.4 to 4.1). After MSE pretraining, the frozen and fully fine-tuned encoders are level (0.0 points, -0.7 to 0.7).

A.5 BCI-IV-2a

BCI-IV-2a contains scalp EEG recorded at Graz University of Technology from nine participants imagining movements of the left hand, right hand, feet, or tongue [12]. We use all 5,184 trials, with five participants for training and two each for validation and testing, following REVE's released split [19]. We apply windowed future prediction (Section 3) either to the training participants' EEG or to a 3,768-hour subset of TUEG, the larger clinical EEG archive from Temple University Hospital [13], before fine-tuning for motor-imagery classification.

Cross-corpus comparison. We use Euclidean alignment to reduce differences between recording sessions, estimating a whitening matrix for each session without using its labels [20]. With this preprocessing, our RQF trained directly on labeled data performs within 0.9 balanced-accuracy points of the models pretrained on BCI-IV-2a or TUEG. Omitting alignment lowers the accuracy of both pretrained models. Published values come from REVE's benchmark. REVE applies Euclidean alignment in its own preprocessing [3], so we compare its result with our aligned results.

Table 9: BCI-IV-2a four-class motor-imagery test balanced accuracy. Published values come from REVE Table 12 [3], which does not report repeat counts. REVE applies Euclidean alignment (EA) in its own preprocessing and does not state the preprocessing behind its baselines. Our models use full fine-tuning and report mean$\pm$std over three seeds; bold marks our strongest configuration, not the best value in the column.

Model	Params	Pretraining data	n	B-ACC
EEGNet	3k	none	n.r.	$0.4482 \pm .0094$
EEGConformer	550k	none	n.r.	$0.4696 \pm .0106$
SPaRCNet	790k	none	n.r.	$0.4635 \pm .0117$
ContraWR	1.6M	none	n.r.	$0.4678 \pm .0125$
CNN-Transformer	3.2M	none	n.r.	$0.4600 \pm .0108$
FFCL	2.4M	none	n.r.	$0.4470 \pm .0143$
ST-Transformer	3.5M	none	n.r.	$0.4575 \pm .0145$
BIOT	3.2M	PREST+SHHS	n.r.	$0.4748 \pm .0093$
LaBraM-Base	5.8M	2,535 h	n.r.	$0.4869 \pm .0085$
CBraMod	4.0M	>9,000 h TUEG	n.r.	$0.5138 \pm .0066$
REVE-Base	69M	61,415 h	n.r.	$0.6396 \pm .0095$
Ours — paired init. (EA), full-FT	113k	none	3	$0.6021 \pm .0249$
Ours — TUEG SSL (cross-corpus), full-FT	113k	3,768 h TUEG subset	3	$0.5495 \pm .0213$
Ours — TUEG SSL (cross-corpus, EA), full-FT	113k	3,768 h TUEG subset	3	$\mathbf{0.6108 \pm .0089}$
Ours — BCI-IV-2a SSL, full-FT	113k	3.2 h BCI-IV-2a train	3	$0.5026 \pm .0148$
Ours — BCI-IV-2a SSL (EA), full-FT	113k	3.2 h BCI-IV-2a train	3	$0.6088 \pm .0308$

B Implementation and reproducibility details

The companion repository is available at https://anonymous.4open.science/r/rqf-ssl-A984.

B.1 Model configurations

Parameter accounting. RQF parameters are complex-valued, so we count each complex parameter as two real scalars. Table 10 applies this convention to the complete downstream models used in the paper. Totals include the per-filter time constants τ (256 scalars for encoders of width 64 and 512 for width 128).

Model	Encoder	Summary/pooler	Readout	Total
Speech classifier (d4/64)	33.7k	—	35.6k	69.3k
EEG windowed (d4/64)	33.7k	16.6k	59.8k	110k
ISRUC epoch-local windowed (d4/64)	33.7k	16.6k	147k	198k
ISRUC contextual windowed (d4/64)	33.7k	16.6k	173k	224k
BCI-IV-2a (d4/64)	33.7k	16.6k	62.6k	113k
iEEG windowed (d4/128)	133k	65.9k	94.1k	293k

Table 10: Real-scalar parameter counts for the downstream models, rounded to three significant digits (k denotes thousands). Components and totals are rounded independently. The ISRUC contextual model adds two one-layer RQFs over the epoch representations.

Multiscale frequency initialization. Deeper RQF layers are initialized at progressively lower frequencies, so the stack begins with filters spanning several temporal scales. In the four-layer encoders used here, the upper frequency limit is multiplied by 1, $1/2$, $1/4$, and $1/8$ from the first to the fourth layer. This schedule affects only the starting values: during training, every layer can learn frequencies throughout the common range $[1/T,\ f_s/2]$.

Shared RQF configuration. All experiments use the same encoder core: a pure RQF stack with zero-order-hold discretization, split-ReLU activations, and a prospective two-tap input. The same stack is applied to every channel. It contains no skip connections, dropout, or normalization between layers. We initialize $\tau = 2\Delta t$; it stays fixed during pretraining and in the TUAB and BrainTreebank evaluations, while full fine-tuning on BCI-IV-2a, ISRUC, and Speech Commands also updates it. We initialize the bandwidth parameter γ in $(0, 1)$ and use a mixing-weight gain of 0.5 at initialization. Table 11 gives the dataset-specific widths, prediction geometry, and training settings for TUAB and BrainTreebank. Input preprocessing is dataset-specific and described below. Layer normalization in the window pooler and downstream readout does not use dataset-wide statistics. Dense targets are standardized separately from a training-only calibration subset.

Coordinate-aware readout. Multichannel tasks use the spatial readout of Section 3, which allows one model to operate on different electrode montages. Each integer-millimetre coordinate indexes three learned 200×32 axis embeddings, covering positions from -100 to $+99$ mm. Their sum passes through a $32 \to 32 \to 16$ MLP to score 16 spatial basis functions across the observed channels. The corresponding value vectors have width 32, giving projections of $64 \to 512$ for EEG and $128 \to 512$ for iEEG. We concatenate the basis-weighted sums with the channelwise mean, standard deviation, and maximum, then summarize these features over time by their mean and standard deviation before classification. These choices produce the readout counts in Table 10.

B.2 Datasets and preprocessing

Published values come from their source papers and were not re-run under our protocol. Each item below ends with the differences that remain; the controlled comparisons share one protocol within each benchmark and are not affected by them.

- **Speech Commands (10-class)** [15]. We retain the 34,975 exactly one-second clips and use a fixed, class-stratified 70/15/15 split at the clip level. Waveforms are sampled at 16 kHz and standardized using training-set statistics.
- **TUAB (EEG, abnormal detection).** We derive the TCP-16 bipolar montage from the 19 referential 10–20 channels and resample it to 128 Hz. No notch filter is applied. Recordings are divided into non-overlapping 100-second segments and normalized independently as described below. Following the BIOT convention [9], we divide the non-test patients 80/20 into training and validation sets with seed 0. The held-out test cohort and its provenance are described in Appendix A. We evaluate on 10-s crops of these segments and on whole 100-s segments. LaBraM, CBraMod, and REVE report 10-s segments at 200 Hz; SAMBA reports 10-s and 100-s segments at 128 Hz.
- **ISRUC-SG1 (EEG, five-class sleep staging).** We use the NeMAR v1.0.1 release and the sleep-stage labels provided by scorer 1. We use six EEG channels (F3–A2, C3–A2, O1–A2, F4–A1, C4–A1, and O2–A1) at 200 Hz and score 30-second epochs [6]. Each recording

is filtered at 0.3–35 Hz with a fifth-order Butterworth filter and at 50 Hz with a notch filter ($Q = 30$), before division into epochs. Subjects 1–80, 81–90, and 91–100 define the training, validation, and test splits; I008 is excluded because F3 and F4 are absent from the released recording. SSL uses only the training subjects. For downstream evaluation, we divide each recording into non-overlapping blocks of 20 consecutive labeled epochs. Unknown-stage epochs interrupt a block, and incomplete blocks are discarded without padding, leaving 87,100 epochs. Both RQF configurations are evaluated on these same retained epochs; only the contextual model uses information from neighboring epochs. The published values use 80 training subjects, a different context module, and parameter counts that omit it; the caption of Table 5 gives the details.

- **BCI-IV-2a (EEG, four-class motor imagery).** We retain all 5,184 trials and use the 22 EEG channels, excluding EOG. Signals are common-average referenced and filtered with a causal fifth-order Butterworth bandpass at 0.3–40 Hz. We extract the four-second cue interval and resample it from 250 to 200 Hz. Subjects 1–5, 6–7, and 8–9 provide 2,880 training, 1,152 validation, and 1,152 test trials, respectively. The downstream readout uses two non-overlapping two-second windows per trial. When Euclidean alignment is used, its transformation is estimated separately from the unlabeled trials of each recording session, including validation and test sessions, and applied to all trials in that session. In-domain SSL sees only training-subject trials. We transfer only the encoder; the downstream learned window pooler and spatial readout are initialized anew. The published values come from REVE Table 12 with REVE's released split; REVE applies Euclidean alignment, and the preprocessing and repeat counts of the other models are not reported.
- **BrainTreebank (iEEG, sentence onset).** We use three-second segments at 2,048 Hz, retaining the channels specified by the clean-channel metadata and excluding contacts labeled GRID or VENT. Signals are Laplacian-referenced using adjacent contacts and notch-filtered at 60 Hz and its harmonics through 360 Hz ($Q = 30$). Each segment is standardized separately within each channel. Electrode coordinates are provided to the downstream spatial readout. We use BaRISTA's chronological fold generator: seven held-out sessions with five folds each. The published values in BaRISTA's Appendix Table 16 use the same sessions and folds and report mean±s.e.m.; they are not paired with our results.

	TUAB	BrainTreebank
Signal	scalp EEG	intracranial EEG
Encoder depth / width	4 / 64	4 / 128
Sampling rate f_s	128 Hz	2,048 Hz
Sequence context T	100 s	3 s
Window / step	1 s / 1 s (100 windows)	62.5 ms / 62.5 ms (48 windows)
Window summary	complex mean $\oplus$ final state $\rightarrow$ linear + LayerNorm	
Summary width	64	128
SSL horizon h_w	1 window (1 s); 2 windows (2 s) for the dense comparison	1 window (62.5 ms)
Inverse temperature β	10	
Sampled negatives	256, sampled uniformly from other window summaries in the batch (any channel)	
Loss weights, global / variance-hinge	1.0 / 0.04	
Pretraining optimizer	AdamW, lr 10^{-4}, weight decay 5×10^{-2}, grad-clip 1.0	
Pretraining epochs / batch	30 / 8	20 / 2
Downstream input / epochs / batch	10-s crops or 100-s segments / 6 / 128 or 16	3 s / 20 / 2 (grad. accum. 2)
Downstream optimizer	AdamW, lr 10^{-3}, wd 0.1 on weight matrices, cosine, weight averaging (Section B.4)	AdamW, lr 2×10^{-4} pretrained and 2×10^{-3} new; warmup-exponential; wd 10^{-2}
Model selection	validation only; test set evaluated once per configuration	

Table 11: Windowed TUAB and BrainTreebank configurations; shared settings span both columns. ISRUC and SC10 use separate schedules (Section B.4), as does BCI-IV-2a.

Segment normalization. To reduce sensitivity to isolated amplitude extremes, TUAB segments are scaled independently in each channel: $x \mapsto x/\big(q_{95}(|x|) + 10^{-8}\big)$, where the percentile is computed over time. This operation neither centers nor clips the signal.

B.3 Pretext construction

Pretext-task geometry. The two objectives use different units of prediction. Windowed prediction groups the encoder states into non-overlapping intervals, whereas dense prediction retains a source at every timestep. In the dense case, N_w in Eq. (3) controls only the interval used to measure future power; it does not pool the source representation.

For a window with sample indices $\mathcal{I}_j$, the mean summary is

$$\bar{\mathbf{t}}_j = \frac{1}{|\mathcal{I}_j|} \sum_{n \in \mathcal{I}_j} [\operatorname{Re} \mathbf{y}_n^{(L)} \,;\, \operatorname{Im} \mathbf{y}_n^{(L)}].$$

Writing $m_j = \max \mathcal{I}_j$ and $\mathbf{s}_j = [\operatorname{Re} \mathbf{y}_{m_j}^{(L)} \,;\, \operatorname{Im} \mathbf{y}_{m_j}^{(L)}]$ for the final state in the window, the learned pooler produces

$$\mathbf{t}_j = \mathrm{LN}(\mathbf{P}[\bar{\mathbf{t}}_j \,;\, \mathbf{s}_j])\,.$$

The parameter-free configuration uses $\mathbf{t}_j = \bar{\mathbf{t}}_j$ directly.

Family	Input $[T \times C]$; rate	Encoder	Windows	Summary	Horizon
Speech dense	$16{,}000 \times 1$; 16 kHz	d4/64	— (N_w=400)	128/64	1,600 smp = 100 ms
Speech windowed (mean)	$16{,}000 \times 1$; 16 kHz	d4/64	40×400	128	4 win. = 100 ms
EEG windowed+pooler	$12{,}800 \times 16$; 128 Hz	d4/64	100×128	64	2 win. = 2,000 ms
EEG dense	$12{,}800 \times 16$; 128 Hz	d4/64	— (N_w=128)	128/64	256 smp = 2,000 ms
iEEG windowed+pooler	$6{,}144 \times C$; 2,048 Hz	d4/128	48×128	128	1 win. = 62.5 ms

Table 12: Pretext geometry used for each family. Windowed horizons are in windows (h_w), dense horizons in samples (Δ). "Summary" gives the dimensionality entering the prediction head (dense: predictor source / target key). The executed EEG windowed-vs-dense pair is horizon-matched at 2 s (2 windows vs 256 samples).

Contrastive objective parameters. Every contrastive configuration uses cosine similarity with inverse temperature $\beta = 10$, 256 sampled negatives, and detached targets. Windowed prediction on TUAB, BrainTreebank, ISRUC, and SC10 samples other window summaries from the batch with replacement. BCI-IV-2a and its TUEG pretraining source instead sample 256 distinct summaries from the same channel in other examples. The source and positive windows themselves never overlap. Single-channel dense prediction excludes nearby anchors whose target intervals overlap, while multichannel dense prediction draws from the same channel in other examples.

Dense targets are standardized coordinate by coordinate using 256 seeded training examples, with a standard-deviation floor of 10^{-6} and $\epsilon = 10^{-6}$ in Eq. (3). The windowed objective adds a variance-hinge penalty of weight 0.04 to discourage collapsed summaries; the dense objective uses only the contrastive loss.

Prediction-horizon selection. The horizon should be long enough that copying the present is ineffective, but short enough that the future remains predictable. For the matched TUAB windowed–dense comparison, a preregistered diagnostic selects the shortest candidate for which a persistence predictor has copy-$R^2 \le 0.50$ while a ridge-affine predictor retains $R^2 \ge 0.20$. On the 100-second recordings, copy-R^2 falls from 0.99 at 125 ms to 0.14 at 2 s, while affine-R^2 remains 0.70 at 2 s; the rule therefore selects 2 s. The headline TUAB configuration retains the earlier one-second horizon. A validation-only comparison on ISRUC likewise selects two one-second windows over one.

The 100-ms speech horizon predates this rule. A later controlled screen found that shortening it to 12.5 ms made the target largely copyable and reduced frozen-encoder accuracy by 7–8 points, supporting the same choice qualitatively.

B.4 Training and model selection

Pretraining schedules. TUAB and BrainTreebank use AdamW with learning rate 10^{-4}, weight decay 5×10^{-2}, and gradient clipping at 1.0; SC10 uses learning rate 10^{-3} and weight decay 10^{-2}. BCI-IV-2a and TUEG pretraining use AdamW with peak learning rate 10^{-4} and weight decay 0.05. The learning rate warms up for five and two epochs, respectively, then follows a cosine schedule ending at 5% of its peak. Weight decay applies to all optimized parameters, including the frequency and bandwidth parameters $(\boldsymbol{\omega}, \boldsymbol{\gamma})$. Downstream evaluation always starts from the final pretraining epoch, so validation performance never selects the SSL checkpoint. ISRUC pretraining runs for 30 epochs on individual 30-second training epochs, using one-second windows and a two-second horizon. SC10 pretraining runs for 50 epochs. For BCI-IV-2a, in-domain SSL runs for 60 epochs using eight non-overlapping half-second windows per trial and a one-second prediction horizon; the cross-corpus models use the final TUEG checkpoint, trained for 30 epochs on a 3,768-hour subset.

Downstream training. Encoder-only transfer reuses the pretrained encoder and initializes the downstream pooler and readout anew. Encoder-and-pooler transfer also reuses the pretrained window pooler. The SSL prediction head is discarded in both cases.

The default fine-tuning settings use AdamW with weight decay 10^{-2} on all optimized parameters, including $(\boldsymbol{\omega}, \boldsymbol{\gamma})$ when trained. Pretrained parameters receive learning rate 2×10^{-4} and newly initialized readout parameters receive 2×10^{-3}; both rates warm up linearly for five epochs and then decay exponentially. The frozen BrainTreebank windowed evaluations keep both the encoder and transferred pooler fixed, optimizing only the spatial readout and classifier at the latter rate. TUAB and ISRUC fine-tune pretrained and paired-initialization models with shared fine-tuning settings: AdamW at learning rate 10^{-3} with weight decay on weight matrices only (0.1 for TUAB, 0.05 for ISRUC), one warmup epoch followed by cosine decay, gradient clipping at 1.0, an exponential moving average of the weights, and channel dropout and gain augmentation. The averaged weights with the best validation AUROC (TUAB) or Cohen's κ (ISRUC) are selected. TUAB trains for six epochs on random 10-s crops of each 100-s segment, or on whole segments for 100-s inputs; frozen TUAB models train only the spatial readout and classifier. ISRUC transfers the encoder alone and trains the pooler, spatial readout, context RQFs, and classifier for 30 epochs with class-weighted cross-entropy. The epoch-local ISRUC comparison and BCI-IV-2a use the default settings above with a 50-epoch cap. SC10 follows its separate community settings for up to 200 epochs: AdamW at 10^{-2} with no weight decay and $\epsilon = 10^{-4}$.

Reporting and model selection. Unless stated otherwise, results are reported as mean $\pm$ standard deviation over three downstream training runs with different random seeds, all starting from one fixed checkpoint. This measures variation in downstream training, conditional on that checkpoint. We select the downstream epoch using validation data and evaluate the test set once at that epoch.

Low-label evaluation. At label budgets of 1, 5, 10, and 25 percent, we construct nested TUAB patient subsets using sampling seed 0, retaining both classes at every budget. The 1% condition contains 16 training patients and 279 100-s segments. Larger budgets retain the previously selected patients; validation and test patients remain unchanged. Each pretrained encoder is compared with its byte-identical initialization before SSL, under both frozen and full fine-tuning, with three downstream seeds per condition.

C Related work

C.1 Predictive representation learning

Contrastive predictive coding (CPC) predicts future latent representations from an autoregressive context [8]. Later speech methods instead mask the latent sequence: wav2vec 2.0 contrasts quantized targets, while HuBERT predicts clustered targets [21, 22]. Joint-embedding predictive architectures predict representations of masked image or video regions using an exponential-moving-average target encoder [23, 24].

Our prediction axis is time along the same trajectory. The windowed InfoNCE configuration is closest to CPC. Its regression variant shares the feature-prediction form used by JEPA, but predicts a later

state of the same encoder instead of using a separate target encoder. Neither configuration masks the input or requires a quantizer.

C.2 Self-supervision with state-space encoders

State-space encoders have also been used with several self-supervised objectives. Audio Mamba reconstructs masked spectrogram patches with input-dependent state transitions and slightly fewer parameters than its matched Transformer [25, 26]. RQF poles are instead input-independent, and the SC10 windows summarize filter states computed directly from the waveform. Related work includes CPC with a structured SSM for ECG, contrastive S4 variants, and predictive, joint-embedding, diffusion, and state-retention objectives for ECG and EEG [27–29, 14, 30–32]. Our scope is the future-state objective for a compact complex-valued RQF; predictive self-supervision with state-space encoders is already established.

C.3 EEG and intracranial foundation models

The EEG references in Figure 2 span several pretraining objectives. LaBraM, CBraMod, and REVE use masked prediction with Transformer encoders; BIOT learns contrastively across datasets; and SAMBA uses masked reconstruction with a Mamba encoder–decoder [1–3, 9, 14]. Their TUAB configurations range from 1M to 369M parameters.

For intracranial recordings, Brant, BrainBERT with a Population Transformer, and BaRISTA provide the 1M–500M parameter references [33–35, 11]. Figure 2 uses BaRISTA's chronological evaluation, with the same sessions and folds as ours.